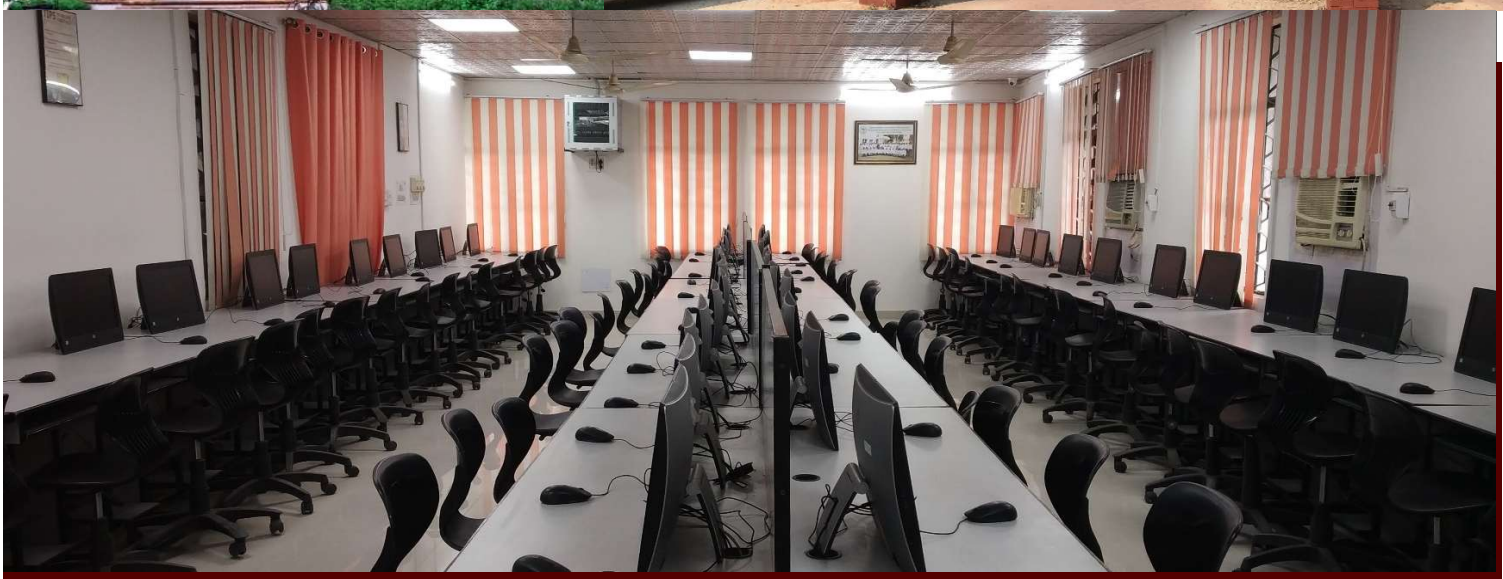
جب کوئی قوم فن اور علم سے عاری ہو جاتی ہے تو وہ غربت کو دعوت دیتی ہے اور جب غربت آتی ہے تو وہ ہزاروں جرائم کو جنم دیتی ہے۔

"When a nation becomes devoid of art and learning, it invites poverty and when poverty comes it brings in its wake thousands of crimes."
-Sir Syed Ahmad Khan

B.Sc. (Hons) (CA) VIII Semester Lab



Laboratory Course-VIII (H)
Course Code - CABSMJ8P01

DEPARTMENT OF COMPUTER SCIENCE

ALIGARH MUSLIM UNIVERSITY, ALIGARH

2025-2026

Credits

The following lab manual up-gradation committee updated the lab manual:

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- ❒ *Prof. Mohammad Ubaid Ullah Bokhari*
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First Edition: December 2025

Department of Computer Science, A.M.U.,

Aligarh (U.P.) India

Lab Manual: Lab – VIII (H) (CABSMJ8P01)

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COURSE TITLE: Lab-VIII (H) **COURSE CODE:** CABSMJ8P01

CREDIT: 4 **PERIODS PER WEEK:** 6

CONTINUOUS ASSESSMENT: 60 **EXAMS:** 40

COURSE DESCRIPTION

This Data Science lab course is designed for students pursuing BSc (Hons) (Computer Science) degrees to gain practical exposure to Data Science concepts, tools, and applications. The course covers fundamental programming concepts, data manipulation techniques, exploratory data analysis, visualization, machine learning implementation, and building end-to-end ML pipelines.

This Web Engineering lab course is designed for students pursuing B.Sc. (Hons) (Computer Science) degrees to develop practical skills in modern web development. The course covers the fundamental technologies and techniques required to build dynamic, interactive, and responsive web applications.

CONTENT

This lab course guides students through Core programming fundamentals using Python, Data structures and manipulation techniques, Numerical computing with NumPy, Data preprocessing and analysis with Pandas, Data visualization techniques, Exploratory Data Analysis (EDA), Machine Learning model implementation and evaluation, Feature engineering and hyperparameter tuning, Building complete ML pipelines from data preparation to model deployment.

This lab Manual also guides students on concepts of Web development using HTML, JavaScript, and PHP

OBJECTIVES

This course is designed to help students in:

- ❒ Use Master Python programming for data science applications
- ❒ Understand and implement data preprocessing and cleaning techniques
- ❒ Implement regression and classification models
- ❒ Evaluate and compare machine learning models
- ❒ Apply feature scaling and hyperparameter tuning
- ❒ Build production-ready ML pipelines
- ❒ Master HTML to create well-structured Web pages.
- ❒ Understand Web Security Fundamentals
- ❒ Build Complete Functional Web Apps.

OUTCOMES

After completing this course, the students would be able to:

- ❒ Write efficient Python programs for data manipulation and analysis
- ❒ Design and style web pages using CSS and layout techniques Handle missing values and outliers appropriately
- ❒ Create insightful visualizations for data storytelling
- ❒ Implement supervised learning models (regression and classification)
- ❒ Implement form validation and user input handling Optimize models using feature engineering and tuning
- ❒ Build and deploy end-to-end machine learning solutions
- ❒ Develop interactive web applications using JavaScript

RULES AND REGULATIONS

Students are required to strictly adhere to the following rules.

- ❒ The students must complete the weekly activities/exercises well in time (i.e., within the same week) that need to be checked and signed by the concerned teachers in the lab in the immediate succeeding week. Failing which activities/assignments for that week will be treated as incomplete.
- ❒ The students must maintain the Lab File of their completed activities/assignments in the prescribed format (Appendix-1).
- ❒ At least **TEN (10)** such timely completed and duly signed weekly activities/assignments are compulsory, failing which students will not be allowed to appear in the final Lab Examination.
- ❒ The students need to submit the following deliverables for each exercise duly signed by the Teacher:
  - Python source code with comments
  - Output screenshots or execution results
  - Brief explanations for approach/methodology
  - HTML/CSS/JavaScript source code with comments
  - PHP source code with database queries
  - Output screenshots showing executed results
- ❒ Late submissions would not be accepted after the due date.
- ❒ The Continuous Lab assessment will be based on two sessional, each carrying 30 marks.
- ❒ Marks distribution for each sessional:
  - 10 Marks: For a duly signed lab report by the respective lab teacher
  - 10 Marks: For solving a lab question/program on the day of the sessional
  - 10 Marks: For the viva conducted during the sessional
- ❒ This distribution ensures a balanced evaluation of students' practical work, problem-solving skills, and conceptual understanding.
- ❒ Cooperate, collaborate, and explore for the best individual learning outcomes, but copying is strictly prohibited.

APPENDIX–I

Template for the Index of Lab File

WEEK NO.	PROBLEMS WITH DESCRIPTION		PAGE NO.	SIGNATURE OF THE TEACHER WITH DATE
1	1#			
	2#			
	3#			
2	1#			
	2#			
	3#			
3	1#			
	2#			
	3#			

Note: The students should use Header and Footer mentioning their roll no. & name in footer and page no in header.

WEEK #1

OBJECTIVES

- To understand the fundamentals of Web Engineering.
- To explore core technologies and tools used in web development

OUTCOMES

After completing this week, the students would be able to:

- Understand Web Engineering concepts and technologies
- Identify and explain front-end and back-end technologies

Problems

1. In your own words, explain Web Engineering.
2. Explore and list core technologies used in Web Engineering, including HTML, CSS, JavaScript, and PHP, and explain the role of each.
3. Identify front-end and back-end technologies used in web development and give examples of each.
4. Explore popular web frameworks or platforms and write short notes on any two.
5. Write a short note on the usage of web technologies in education, business or social platforms

WEEK #2

OBJECTIVES

- To understand the importance and applications of Data Science
- To identify various roles and frameworks in Data Science

OUTCOMES

After completing this week, the students would be able to:

- Explain Data Science importance in modern applications

PROBLEMS

1. In your own words, explain what Data Science is and why it is important in today's world.
2. Identify and describe any five real-world datasets.
3. List major tools used in Data Science (programming languages, libraries, platforms) and state their purpose.
4. Explore and write short notes on any three Data Science frameworks.
5. Identify different roles in Data Science (e.g., Data Analyst, Data Scientist, ML Engineer) and their responsibilities.
6. Write a short note on the usage of Data Science being in your own domain of interest.

WEEK #3

OBJECTIVES

- To master basic Python and HTML programming fundamentals
- To understand conditional statements and loops
- To work with strings and basic operations
- To work with lists and tables

OUTCOMES

After completing this week, the students would be able to:

- Write Python programs using conditional statements and Apply various HTML tags correctly
- Implement iterative logic using loops
- Manipulate strings and perform basic computations

PROBLEMS

1. Write a Python program to read a number and check whether it is even or odd.
2. Write a Python program to find the largest of three numbers using conditional statements.
3. Write a Python program to generate the first n Fibonacci numbers using a loop.
4. Write a Python program to count the number of vowels in a given string.

Problems Based on HTML

5. Create a HTML page with the following:
   - Webpage Title " My first web page".
   - Add BGCOLOR, TEXT attributes to the body.
   - Show the use of different heading tags.
   - Add two paragraphs to your webpage.

   Show these text with following formatting along with line break:
   - Italics – "This text is italics"
   - Bold – "This text is bold"
   - Underline – "This text is underline".
6. Create a web page to show this output:
   1. List of fruits:
      - Apple
      - Banana
      - Guava

2. List of Vegetables:
   - Carrot
   - Cabbage
   - Spinach

7. Create the following table in HTML with Dummy Data:

Name of the Train	**Place**	**Destination**	**Train No.**	**Time**		**Fair**
				Arrival	Departure	

WEEK #4

OBJECTIVES

- ❒ To understand and work with Python data structures
- ❒ To implement functions and write reusable code
- ❒ To manipulate lists, dictionaries, and other collections

OUTCOMES

After completing this week, the students would be able to:

- ❒ Use lists and perform operations on them
- ❒ Create and use dictionaries for data storage
- ❒ Write and use functions effectively
- ❒ Implement string and collection manipulation techniques

PROBLEMS

1. Create a list of numbers and find the sum and average of the list.
2. Write a function to check whether a given string is a palindrome.
3. Create a dictionary of student names and marks. Display students scoring above 75.
4. Write a program to sort a list in ascending and descending order.

Problems based on HTML

5. Write HTML code to generate the following output:

<table>
<tr><td>1</td><td>2</td><td>3</td><td>4</td></tr>
<tr><td>5</td><td colspan="2" rowspan="2">Image</td><td>6</td></tr>
<tr><td>7</td><td>8</td></tr>
<tr><td>9</td><td>10</td><td>11</td><td>12</td></tr>
</table>

6. Write an HTML code to develop a Web page having two frames that divide the Web page into two equal rows and then divide the second row into two equal columns, then fill each frame with a different background color.
7. Design an HTML Page having 3 images placed in the following format. (Hint: Figure 1)

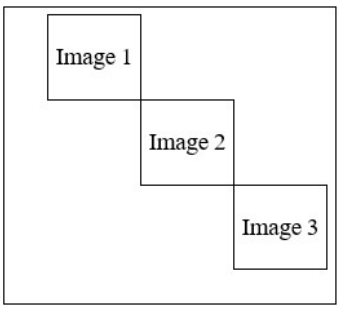

Figure 1

WEEK #5

OBJECTIVES

- To understand NumPy arrays and their properties
- To implement navigation between multiple pages
- To work with array indexing and slicing
- To create interactive HTML forms

OUTCOMES

After completing this week, the students would be able to:

- Create and manipulate NumPy arrays
- To implement navigation between multiple pages
- Use array indexing and slicing effectively

PROBLEMS

1. Create a NumPy array and display its shape, size, and data type.
2. Perform at least five basic arithmetic operations on NumPy arrays (addition of two arrays, subtraction, etc.).
3. Create a NumPy array from having numbers between 1 to 100 and print only the prime numbers in them.
4. Take a NumPy array and demonstrate indexing and slicing on it.

Problems based on HTML

5. Write the code to develop a Web page, as shown below, using frames:

<table>
<tr><td rowspan="2">My Frames</td><td colspan="2"></td></tr>
<tr><td></td><td></td></tr>
</table>

6. Design a page with a text box called ‘name’ and a button with label ‘Enter.’ When you click on the button, another page should open, with the message “Welcome <name>”, where name should be equal to the name entered in the first page. Set default value of ‘name’ text box to Victoria. Add another button called Reset on click of this button name ‘text box’ should be set to 100 default values.
7. Design a form in HTML using all input types.

WEEK #6

OBJECTIVES

- To compute statistical measures using NumPy
- To implement event handlers
- To work with array reshaping and transformations
- To add interactivity using JavaScript

OUTCOMES

After completing this week, the students would be able to:

- Calculate mean, median, and standard deviation
- Perform matrix addition and multiplication
- Reshape arrays to different dimensions

PROBLEMS

1. Create a NumPy array and compute mean, median, and standard deviation of its data.
2. Perform matrix addition and multiplication using NumPy.
3. Generate a random array and find its maximum and minimum values.
4. Reshape a one-dimensional array into a matrix of different sizes

Problems based on HTML and JavaScript

5. Design a Web Page, which is like 'compose' page of e-mail

To

Copy

Message

SEND

6. Design a series of three HTML Pages for ABC.COM, each called from the previous one. Accept Name on the first page. When the user clicks on the enter button, second page should open. The second page should not display the name but a 'Welcome screen with some information about ABC.COM. When the user will click on the 'next' button it should display the name accepted in page 1 on page 3. (Hint: you may use hidden fields)
7. Write the segment of Script that would ask the user if he wants a greeting message and if he does, display a Gif file called Welcome.gif and display "Welcome to Department of Computer Science!" in the document window following the Gif.

WEEK #7

OBJECTIVES

- To create and manipulate Pandas Series and DataFrames
- To perform basic data frame operations
- To identify and handle missing values in datasets

OUTCOMES

After completing this week, the students would be able to:

- Create Pandas Series and DataFrames
- Load data from CSV files
- Access and explore DataFrame structure
- Design comprehensive forms with multiple input types

PROBLEMS

1. Create a Pandas Series from a list and a DataFrame from a dictionary.
2. Load a CSV file and display the first five rows and display head, tail, and info of a dataset.
3. Take a DataFrame and do the following:
   - What is the average value of the second column?
   - What is the average value of the first 5 rows of the third and fourth columns?
   - Compute the row wise sum of all possible values in an array.
   - Find the maximum of average value in each row.

Problems Based on HTML and JavaScript

4. Write a program to display a multiplication table.
5. Create a Web page using two image files, which switch between one another as the mouse pointer moves over the image. Use the On Mouse over and On Mouse out event handler.
6. Use the date function gets Date & set Date to prompt the user for an integer between 1 – 31 & return day of the week it represents.

WEEK #8

OBJECTIVES

- To rename columns and modify data types
- To filter data based on conditions
- To design complex HTML forms with JavaScript integration
- To develop advanced web page enhancements

OUTCOMES

After completing this week, the students would be able to:

- Identify missing values and their patterns
- Create interactive web pages with advanced effects
- Modify DataFrame structure and data types
- Filter and subset data efficiently

PROBLEMS

1. Read a dataset and identify the information about the missing values in a dataset.
2. Handle missing values using mean/median/other approaches.
3. Rename columns of the datasets and change their data types.
4. Filter rows and columns based on specific conditions (such as remove rows with missing values, if value is less than 50 etc.).

Problems based on HTML and JavaScript

5. Display time and print message accordingly e.g., 'Good Morning' in Morning etc.
6. Using JavaScript create a digital clock.
7. Design a form as shown below:

Details

Your Name:

Your Address:

Your Gender: Male ○ Female ◉

Your Country: India

Your Opinion:

WEEK #9

OBJECTIVES

- To perform comprehensive exploratory data analysis
- To identify and handle outliers
- To compute correlations between features

OUTCOMES

After completing this week, the students would be able to:

- Generate and interpret summary statistics
- Identify correlations and remove highly correlated features
- Detect and handle outliers appropriately
- Perform complete EDA workflow

PROBLEMS

1. Read a dataset and compute summary statistics of numerical columns.
2. Find correlations between numerical feature and remove highly correlated columns.
3. Draw boxplots, identify outliers and remove them.
4. Perform all EDA operations on a dataset.

Problems based on HTML and CSS

5. Create a website that has a questionnaire (any), on submitting, it displays the inputted data on the other page. You are supposed to use CSS (wherever necessary).
6. Create a website that divides the Web page into two unequal frames. In Frame One, there are two links to two different forms. The forms are validated on submitting and the result is shown at the bottom of the same page. Use CSS for formatting.

WEEK #10

OBJECTIVES

- To create various types of plots using Matplotlib and Seaborn
- To visualize feature relationships
- To generate correlation heatmaps

OUTCOMES

After completing this week, the students would be able to:

- Create line plots, bar plots, and histograms
- Generate boxplots and scatter plots
- Interpret correlation heatmaps
- Create publication-quality visualizations

PROBLEMS

1. Read a dataset and create different line and bar plots using Matplotlib.
2. Use the same dataset and plot histograms and boxplots using Seaborn.
3. Create scatter plots for feature relationships and generate a correlation heatmap

Problems based on HTML and JavaScript

4. Develop a Web page with the following enhancements: • A rollover effect, where an image changes if the user places the mouse over it.
   - An animation that occurs in response to the user clicking on an image.
   - A pull-down menu with each option linking to a specific page.
5. Display the calendar using JAVA SCRIPT code by getting the year from the user.

WEEK #11

OBJECTIVES

- To implement linear regression models
- To split data into training and testing sets
- To evaluate regression model performance
- To implement classification models

OUTCOMES

After completing this week, the students would be able to:

- Prepare data for model training
- Implement linear regression from scratch or using Scikit-learn
- Evaluate regression models using appropriate metrics
- Implement basic classification models

PROBLEMS

1. Read a dataset and split a dataset into training and testing sets.
2. Implement a simple linear regression model.
3. Evaluate regression performance using MAE and RMSE. and visualize actual vs predicted values.
4. Change the regression model and compare the performance with the previous model.

Problems Based on HTML, Java Script and XML:

5. Create a HTML registration form and to validate the form using java script code.
6. Create a HTML file to open new window from the current window using JavaScript.
7. Create an HTML page with 2 combo box populated with month & year, to display the calendar for the selected month & year from combo box using JavaScript.

WEEK #12

OBJECTIVES

- To evaluate classification performance
- To compare multiple classification algorithms
- To work with XML and JavaScript.
- To create dynamic web pages

OUTCOMES

After completing this week, the students would be able to:

- Generate and interpret confusion matrices
- Calculate and interpret performance metrics
- Compare different classification algorithms

PROBLEMS

1. Read a dataset and implement a basic classification model.
2. Generate and interpret a confusion matrix. Calculate accuracy, precision, recall, and F1-score and visualize classification results.
3. Use various classification models and compare them on the same datasets

Problems Based on HTML and XML:

4. Create a CD catalog using XML file.
5. Create external style sheet and using the style sheet in XML file.
6. Create an XSL style sheet to display the data in the xml using HTML table.

WEEK #13

OBJECTIVES

- To apply feature scaling techniques
- To detect and prevent overfitting
- To perform hyperparameter tuning
- To build end-to-end machine learning pipelines
- To perform comprehensive EDA on application-based datasets

OUTCOMES

After completing this week, the students would be able to:

- Apply standardization and normalization techniques
- Identify overfitting and adjust models accordingly
- Perform server-side form validation
- Compare model performance with and without scaling

PROBLEMS

1. Read a dataset and apply feature scaling techniques.
2. Demonstrate overfitting using scores and adjust the model, if required.
3. Apply basic hyperparameter tuning and compare it with default setting.

Problems Based on HTML, Java Script, XML and PHP:

4. Create a PHP program to demonstrate the different predefined function in array, Math, Data & Regular Expression.
5. Create a PHP file that contains one textbox for Salary and one Submit button. If user provides salary and submits the button, then name of such employees should be displayed, who are having salary less than Salary value supplied by the user. Write prepared statements for database query. (Note: Schema of Employee table is Employee (EmployeeID, Name, Salary, DepartmentID)).

WEEK #14

OBJECTIVES

- To integrate all learned concepts
- Use regular expressions in PHP
- To perform form validation on the server side
- To work with prepared statements

OUTCOMES

After completing this week, the students would be able to:

- Execute complete machine learning workflows
- Write PHP code to connect to databases
- Perform server-side form validation
- Deploy simple ML models

PROBLEMS

1. Perform EDA on an application-based dataset.
2. Build an end-to-end ML pipeline.

Problems Based on HTML and PHP:

3. Design a page that contains four textboxes for Name, Email ID, Age and Address and one Submit button. Once user supplied these values a server-side validation should be performed and data should be inserted into the table Employee (Name, Email ID, Age, Address)).